Harish Venkatesan M R

Robotics Engineer | ROS 2 | Autonomous Systems | Computer Vision
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Right to work: UK Graduate visa, full-time work permitted until January 2027 (Skilled Worker sponsorship required thereafter)

PERSONAL PROFILE

Robotics engineer with an MSc in Intelligent Robotics from the University of York and 18 months of commercial software engineering experience. Builds autonomous robots end to end, from sensors and embedded control on Arduino and Raspberry Pi to ROS 2 perception, SLAM and Nav2 navigation in Gazebo. Led a 4-person team to deliver a rescue robot with a custom YOLOv5 model achieving 86% mAP. Seeking a Robotics, Automation or Controls Engineer role in the UK.

KEY SKILLS

- **Robotics Software:** ROS 2 (Jazzy), Nav2, SLAM Toolbox, Gazebo, TurtleBot3, rclpy, URDF/xacro, RViz
- **Autonomy and Control:** autonomous navigation, SLAM, obstacle avoidance, sensor fusion, state machines, PID/proportional control, closed-loop tuning, kinematics and dynamics, Mecanum and differential drive
- **Embedded and Sensors:** Arduino, Raspberry Pi, UART serial, I2C, IR and ultrasonic sensors, encoders, cameras
- **Computer Vision:** YOLOv5 custom training, Ultralytics YOLO, OpenCV, PyTorch
- **Programming and Tools:** Python, C/C++, Java, Git, Linux (Ubuntu), MATLAB, Simulink, Autodesk Inventor, Fusion 360
- **Engineering Practice:** system integration, test and validation, hardware-software debugging, code review, technical documentation, Agile/Scrum, team leadership

ROBOTICS PROJECTS

Autonomous Search and Rescue Robot University of York, 2024

Team Lead (team of 4) - Python, C++, Arduino, Raspberry Pi, YOLOv5, OpenCV | Demo: youtube.com/watch?v=4F5axwGFdGk

- Led a team of 4 engineers, assigning subsystem ownership across ground and aerial platforms and integrating all workstreams into a single working system for disaster response.
- Delivered autonomous maze navigation for a Mecanum-wheeled robot, fusing IR and ultrasonic sensor data for obstacle avoidance, with motor control on Arduino and high-level logic on Raspberry Pi in Python.
- Trained a custom YOLOv5 model to detect victims and obstacles, achieving 86% mAP, and fed live detections directly into navigation decisions.
- Developed the serial communication layer linking the sensor array to a custom two-jaw gripper, enabling the robot to grasp a rescue object and return it to the start position without operator input.

ROS 2 Autonomous Navigation (SLAM + Nav2) Personal project

Python, ROS 2, SLAM Toolbox, Nav2, Gazebo, TurtleBot3

- Mapped an unknown simulated environment with a TurtleBot3 using SLAM Toolbox, then navigated autonomously to goal poses on the saved map with the Nav2 stack.
- Wrote a custom Python ROS 2 node that sends a sequence of waypoints to Nav2 so the robot patrols a route autonomously.

Vision-Based Target Following Robot (ROS 2 + Gazebo) Personal project, 2026

Python, ROS 2 Jazzy, Gazebo Harmonic, OpenCV, YOLO

- Designed a ROS 2 system in which a camera-equipped differential-drive robot detects a target and drives to it, with perception and control in separate nodes so the same pipeline runs in simulation and on a webcam.
- Built a detector node with two modes, CPU-only HSV colour segmentation and YOLO (Ultralytics) detection of any COCO class, publishing target offset, size and confidence.
- Implemented a SEARCH / TRACK / ARRIVED state-machine controller with proportional steering and distance control at 20 Hz, using velocity limits, deadband and lost-target timeout to prevent oscillation.
- Modelled a custom robot in URDF/xacro with a Gazebo camera and DiffDrive plugin, bridged to ROS 2 through ros_gz_bridge, with all gains exposed as ROS parameters for live tuning.

PROFESSIONAL EXPERIENCE

Associate Engineer - Technology, Virtusa Consulting Services 05/2022 - 09/2023

Chennai, India

- Software test engineer in a 10-15 person Agile/Scrum team delivering enterprise applications built on Java, Spring Boot, REST APIs and React.js, across the full development lifecycle.
- Designed and executed unit and integration tests for backend services, REST APIs and front-end features, improving the reliability of production releases.

- Debugged defects across backend and front end: reproduced issues, isolated root causes and verified fixes before release.
- Carried out peer code reviews focused on correctness, test coverage and maintainability.
- Wrote and maintained test and technical documentation, and supported CI/CD pipelines, AWS deployments and release management.

Intern - Delivery, Virtusa Consulting Services 01/2022 - 03/2022
Chennai, India

- Worked on Java and Spring Boot backend tasks, testing and code reviews; converted to a full-time Associate Engineer role on graduation.

Robotics Team Member, SP Robotics 2018

- Taught practical robotics to students alongside senior staff over 6 months.

EDUCATION

MSc Intelligent Robotics, University of York 09/2023 - 01/2025
Modules: Robot Kinematics and Dynamics; Control and Instrumentation for Robotics; Practical Robotics; Embedded Systems Design and Programming; Machine Learning

BE Computer Science and Engineering, KPR Institute of Engineering and Technology, India 08/2018 - 05/2022
First Class. Modules: Data Structures; Object-Oriented Programming; Computer Networks; Human-Computer Interaction; Natural Language Processing

ACHIEVEMENTS AND CERTIFICATIONS

- Grand Finalist, Virtusa NeuralHack Season 5 (2021): top 250 of 100,000+ participants in a national hackathon; earned a pre-placement offer.
- Oracle Certified Associate, Java SE 8 Programmer - Oracle (2022)